The 《Taichu》 and 《Santong》 Calendars: Errors in the Synodic Month and the Intercalation Cycle

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Abstract

In the first year of Taichu (104 BCE) the Han court replaced its calendar. The transmitted record of that reform is thin on ritual and rich on arithmetic: what survives is a table of constants in the Book of Han, under the heading 「统母」, the "mothers" from which every later figure was derived. Two of those constants govern everything that follows. One fixes the length of the month; the other fixes how many intercalary months fall in how many years. This essay reads both, and then reads the record of what happened to them. For a hundred and some years the calendar ran; then the histories begin to say that the new moon was arriving before the calendar said it would, that the first day of the month was being written on a day when no moon could be seen. At Yuanhe 2 (85 CE) the court conceded that the calendar had drifted a full day from the sky. The evidence it used was not a new observation but an old one: the solar eclipses already recorded in the histories, sorted by whether the calendar had put them on the first day of the month. The essay follows that audit. Its conclusion is narrow. The constants of the Taichu reform were internally consistent and were the best available at the time; they were also, by construction, too coarse, and the record says so in its own words.

Keywords: Taichu calendar; 太初历; Santong calendar; 三统历; synodic month; 朔策; intercalation cycle; 闰周; 19-year cycle; Book of Han; 汉书·律历志; Book of the Later Han; 后汉书·律历志; Jia Kui; 贾逵; solar eclipse; Chinese calendrics

Introduction

In 104 BCE the Han court changed its calendar. The histories record the change as a matter of ritual and of dynastic colour: a new first month, new robes, a new reign title. What they do not

record is how the calendar was computed.

That part survives elsewhere, in a table. The Book of Han, in its treatise on harmonics and the calendar, lists the constants under the word 「统母」 — the mothers of the system. Every later quantity in the calendar was generated from them. Seven of them are given in a single passage, and two of those seven decide whether the calendar will hold.

The first is the length of the month. The second is the intercalation cycle — how many extra months are inserted in how many years.

Both were fixed by decree of arithmetic, not by measurement alone. Both were, in the event, slightly wrong. This essay is about how slightly, and about how the wrongness was found out.

The finding out is the interesting part. It was not done by a philosopher. It was done by clerks comparing two lists: the dates on which the calendar said the moon would be dark, and the dates on which the histories said the sun had been eclipsed. An eclipse can only happen at new moon. So every recorded eclipse is a free test of the calendar's new moon, and the histories of the Han had been recording eclipses for two centuries.

1. The reform of Taichu changed a ruler, not just a year

Qin and the early Han had used the Zhuanxu calendar, with the year beginning in the tenth month. The Han inherited it and did not change it, down to the reign of Emperor Wu.

At Taichu 1 the court changed it. The first act was to move the beginning of the year back to the first month. That has been described in an earlier essay in this series, on the Zhuanxu calendar and its archaeological evidence, and is not repeated here.

The second act is less often noticed, and it matters more for what follows. The reform did not only move the start of the year. It redefined the grain into which a day was divided.

The calendar-makers of the Han did not work in decimals. They worked in fractions. The number of parts into which one day was cut was called the 「日法」, the day-divisor. Once that divisor is fixed, every month and every year must land on one of those parts. A month cannot be 29.53 days; it can only be some whole number of eighty-firsts of a day.

The Taichu reform set the day-divisor at eighty-one.

Eighty-one was not an arbitrary choice, and the Book of Han says where it came from in the first line of the table:

> 日法八十一。元始黄钟初九自乘，一龠之数，得日法。

Eighty-one is nine multiplied by nine. Nine is the yang line of the Changes; the standard pitch-pipe, the Yellow Bell, holds one yue, and one yue is eighty-one parts. So the divisor was derived from music theory and from divination, not from the sky.

This is the point that governs the rest of the essay. A calendar must agree with heaven, but the ruler in the calendar-maker's hand was first a numerical and cosmological ruler. The error that appears a century later has its root here.

2. 《Taichu》 and 《Santong》: one set of numbers, two names

There is a confusion worth clearing up at the start. The calendar introduced in the Taichu era is called the Taichu calendar. The complete algorithm preserved in the Book of Han is called the Santong calendar, the calendar of the Three Sequences. Two names, one set of numbers.

The Book of the Later Han, looking back, uses the second name even for the first year:

> 自太初元年始用《三统历》，施行百有余年，历稍后天，朔先于历，朔或在晦，月或朔见。

"From the first year of Taichu the Santong calendar was in use." Later writers simply called the whole system by the name it acquired. The name is associated with Liu Xin, who at the end of the Western Han reorganised the calendar and dressed its numbers in a scheme of three sequences — heaven, earth, and man; black, white, and red — mapped onto the three dynasties of Xia, Shang, and Zhou. The numbers were unchanged. The vocabulary was new.

So when a Han source sets 「太初」 and 「三统」 side by side, they are not two calendars. The difference is in the name, not in the arithmetic.

3. The synodic month: twenty-nine days and forty-three eighty-firsts

The table of constants in the Book of Han gives everything at once:

> 日法八十一。……月法二千三百九十二。……闰法十九，因为章岁。……章月二百三十五。……统法千五百三十九。……周天五十六万二千一百二十。……朔望之会百三十五。

Seven constants, each with a use. The first two govern the length of the month: the day-divisor is eighty-one, and the month-factor is two thousand three hundred and ninety-two.

The day-divisor cuts the day into eighty-one parts. The month-factor says how many such parts a month contains: two thousand three hundred and ninety-two. Read against the divisor, that is twenty-nine whole days and forty-three of the eighty-one parts left over. This is not a modern reconstruction; it is how Han readers read the table. A month runs a little over twenty-nine and a half days, and the half-day's remainder is written as forty-three eighty-firsts.

Why not divide further? Because the divisor had already been fixed at eighty-one. The denominator is chosen first; the grid is then fixed; and the moon is required to fall on the nearest grid line.

That remainder of forty-three eighty-firsts became the signature of Han calendrics, and gave the system its common name: the eighty-one-part method. The calendar-makers were proud of their denominator. The Book of Han puts it first in the table and explains, at length, why it descends from the Yellow Bell.

4. The intercalation cycle: seven months in nineteen years

The second constant is the intercalation cycle.

Twelve lunar months are shorter than a solar year. The shortfall accumulates; every two or three years it amounts to a whole month, and a month must then be inserted. How many insertions in how many years is the great joint of any lunisolar calendar: too many and the seasons run ahead of the months, too few and they fall behind.

The Han solution was seven intercalary months in nineteen years. Two entries in the table state it: the intercalation divisor is nineteen, which is also the length of a zhang; and the months in a zhang number two hundred and thirty-five.

Where two hundred and thirty-five comes from is spelled out in the Han commentaries: nineteen years of twelve months gives two hundred and twenty-eight; add seven intercalary months and the total is two hundred and thirty-five. One zhang is nineteen years and two hundred and thirty-five months, containing seven intercalations.

The reasoning behind the cycle is that nineteen tropical years are approximately equal to two hundred and thirty-five synodic months. Approximately, not exactly. The Han astronomers knew this, which is why they called the period a zhang — a chapter, a unit that closes and then reopens. Within one chapter the new moon and the winter solstice return to the same day. The remainder left over at the end of the chapter does not vanish; it accumulates into the next.

The Book of Han also lists a further constant, 「朔望之会百三十五」, and a further period, the hui-month of six thousand three hundred and forty-five, precisely in order to handle that accumulating remainder. Eclipses occur at new and full moon; if the new moons are off, the predicted eclipses are off. The periods were stacked — zhang, tong, yuan — in the hope of absorbing the remainder entirely.

They were not absorbed. That is the subject of the next section.

5. When the error became visible

The Taichu calendar ran for something over a century. Then the record says it had drifted.

The Book of the Later Han states the matter first in summary:

> 自太初元年始用《三统历》，施行百有余年，历稍后天，朔先于历，朔或在晦，月或朔见。

The calendar was "slightly behind heaven": the sky had reached a position the calendar had not yet reached. The new moon "came before the calendar": the calendar still showed the old month while the sky had already turned to the dark of the moon. And then the plainest sentence in the passage: the first day of the month was sometimes written on a day of true darkness, and a crescent was sometimes visible on the day the calendar called the dark of the moon.

These are not minute quantities. They are things a person can see.

By Yuanhe 2, under Emperor Zhang, the gap had grown:

> 至元和二年，《太初》失天益远，日、月宿度相觉浸多，而候者皆知冬至之日日在斗二十一度，未至牵牛五度，而以为牵牛中星，后天四分日之三，晦朔弦望差天一日，宿差五度。

At the winter solstice the sun should have been in the twenty-first degree of the Dipper; the calendar still had it at the determinative star of the Ox. New moon, first quarter, full moon, last quarter — the calendar was a full day out, and the lunar lodges five degrees out. And "the observers all knew it": this was not the discovery of one clever man but the common knowledge of the men who did the watching.

The emperor issued an edict. One sentence in it names the fault precisely:

> 史官用太初邓平术，有余分一，在三百年之域，行度转差，浸以谬错。

The system of Deng Ping, used by the astronomers, "carries a remainder of one"; the remainder is small, and within the span of three hundred years it turns into a drift. The edict also quotes a prognostic text: 「三百年斗历改宪」 — in three hundred years the calendar of the Dipper is reformed. The line reads as prophecy and functions as retrospection.

The reform that followed adopted the Quarter-Day calendar. But the argument did not end. Bian Xin and Li Fan proposed that the first month of the reckoning should begin with a long month, so that the quarters would fall on fixed days. Their opponents pointed at the intercalation cycle itself:

> 而訢、梵犹以为元首十一月当先大，欲以合耦弦望，命有常日，而十九岁不得七闰，晦朔失实。

"Nineteen years no longer yields seven intercalations." The cycle itself had stopped closing. Once the cycle fails, the new moons fail with it. The dispute was settled by imperial ruling: the initial month of the epoch was to be a short month.

6. Checking the calendar against eclipses

The hardest document in the dispute is an audit by Jia Kui.

Jia Kui took the new moons computed by the Taichu calendar and set them against the solar eclipses recorded in the histories. An eclipse can occur only at new moon. Every recorded eclipse is therefore a ready-made test of the calendar's new moon. He counted in three blocks:

> 以《太初历》考汉元尽太初元年日食二十三事，其十七得朔，四得晦，二得二日；……以《太初历》考太初元年尽更始二年二十四事，十得晦；……以《太初历》考建武元年尽永元元年二十三事，五得朔，十八得晦。

In the first block, from the first year of Han to Taichu 1, twenty-three eclipses: seventeen fell on the computed new moon. In the second, from Taichu 1 to Gengshi 2, twenty-four eclipses: ten fell on the dark of the moon — that is, a day late. In the third, from Jianwu 1 to Yongyuan 1, twenty-three eclipses: only five fell on the new moon, eighteen on the dark.

Set the three blocks side by side and the trend is not in doubt. The same calendar, tested by the same rule, agrees with the sky in its early years and disagrees in its later ones.

The method deserves a word of its own. Jia Kui did not ask how many quarter-hours the calendar was out. He asked only whether the eclipse fell on the computed first day of the month. That was the test the Han could actually run: they had no records precise to the minute, but they had two centuries of entries reading "in such a year, such a month, on such a day, the sun was eclipsed." Set that line against the calendar, and either it matches or it does not.

It is worth adding that Jia Kui ran the same test with the newly adopted Quarter-Day calendar and reported those results separately. That is how the argument was actually conducted: not by comparing the elegance of the theories, but by counting which one matched more often.

7. Why the eighty-one-part method drifted

Back to the root.

The fault of the eighty-one-part method is not that eighty-one is the wrong number. It is that eighty-one is too coarse. Cut a day into eighty-one parts and each part is a substantial interval. A month must fall on one of those parts, and the nearest part need not be the moon's actual position.

The modern value for the synodic month falls between two of the grid lines. The eighty-one-part method takes the nearer one, twenty-nine days and forty-three eighty-firsts. A single grid-step looks small. But a small error in one month is twelve small errors in a year, and twelve small errors a year for a century and a half moves the calendar by a whole day. That is exactly what the Book of the Later Han describes when it says the new moon came before the calendar, and that the first of the month was sometimes written on a day of true darkness.

The intercalation cycle fails for the same kind of reason. Seven months in nineteen years treats nineteen tropical years and two hundred and thirty-five synodic months as equal. They are close, not equal. The difference accumulates year by year until, as the record puts it, nineteen years no longer yields seven intercalations and the cycle will not close.

The two faults run in different directions. A slight error in the month length shifts the new moons gradually. A slight error in the cycle gradually detaches the seasons from the months. Together they throw the year, the month, and the day out of joint at once.

The Quarter-Day calendar that replaced it cut the day into four parts and wrote the month as twenty-nine days and four hundred and ninety-nine nine-hundred-and-fortieths. A larger denominator means a finer grid and a closer fit. But it is still a fraction chasing the moon. That approach lasted until the Tang, when Yixing introduced an algorithm of second differences for unequal intervals.

8. What can be checked, and what is only inference

Two sides, as usual.

What can be checked: the seven constants of the table, written out in the Book of Han — day-divisor eighty-one, month-factor two thousand three hundred and ninety-two, intercalation divisor nineteen, months in a zhang two hundred and thirty-five, tong divisor one thousand five hundred and thirty-nine, circuit of heaven five hundred and sixty-two thousand one hundred and twenty, and the syzygy meeting of one hundred and thirty-five. The seven interlock; they are not a later insertion. The year of the reform, the wording of the edict, the dispute of Bian Xin and Li Fan, and the three blocks of Jia Kui's eclipse count are all in the Book of the Later Han and can be checked line by line.

What cannot be checked: the observations made at the time of the reform. What Luo Xiahong and Deng Ping measured, how often, and with what error, the histories do not record. What is known is the outcome — that they chose eighty-one, chose seven in nineteen, chose those fractions. Why those numbers rather than others, apart from the Yellow Bell explanation the Book of Han itself gives, there is no evidence.

【One reading】 That the divisor eighty-one was taken from the pitch-pipes is a self-justification internal to numerology, not an astronomical reason. The calendar-makers first found a handsome number in the numerical arts, and then required the calendar to accommodate it. This reading explains the arrangement of the table, but there is no direct evidence that anyone reasoned in that order.

One further distinction should be kept. When this essay says the calendar was "off", it means off by present-day measurement. In the Taichu years nobody thought so. They thought the new calendar was far better than the Zhuanxu calendar it replaced, and they were right. The drift became visible only after a century and a half. To project the later error backwards and say that the men who made the calendar got it wrong is not a fair reading of what they did.

Note on the text

This is a popular exposition, and it makes no claim to original research. The calendrical facts cited rest on transmitted texts and on excavated material. This English version is issued alongside the Chinese original as the same article in another language. The author's name is 邝楚嘉, Chujia Kuang; he writes under the pen-name 嘉言一得 (Jiayan Yide).

Chinese terms are left in their original characters where the word itself is the subject of the discussion.

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